ESL 3

The Elements of Statistical Learning

Linear Regression

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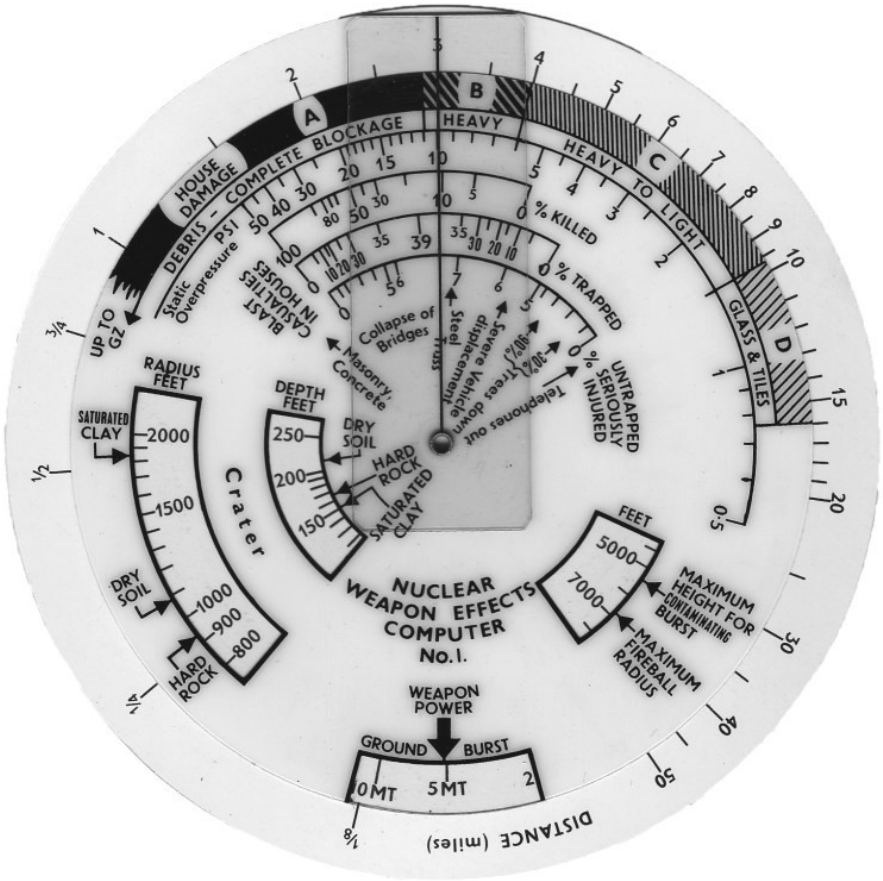
ESL 3: Overview

ISL Book Chapters:	1 and 7
Synopsis:	Linear regression may seem simple, but it forms the foundation for many modern statistical learning methods. This module reviews the core ideas behind linear regression and least squares fitting, then introduces extensions that allow non-linear relationships between predictors and the outcome.
Competencies:	• Understand how least squares is used to fit a regression line. Interpret coefficients, fitted values and residuals in human language, and assess model adequacy through residual diagnostics. • Apply transformations, interactions, and polynomial terms to handle non-linear relationships and know when such extensions are appropriate. • Recognize how these extended linear models relate to more flexible modern methods and identify situations where those methods are preferable.
Prerequisites:	Understanding of probability and statistics is needed at the basic level. Further, limited working proficiency in Tidyverse dialect of R programming language is needed to complete practical exercises.
Tutorials:	
Data:	The Advertisement dataset (ISLR2 package) FIFA22 dataset from Kaggle (https://www.kaggle.com/datasets/stefanoleone992/fifa-22-complete-player-dataset)

Fisher Sliding Rule

Rise of classical statistics and experimental design.
1950s-1970s



A slide rule embeds mathematical relationships directly into its physical scales.

Statistics mirrors this idea in an abstract way: it represents relationships between variables symbolically or computationally. Both are predictive models. Fisher's slide rule is simply a fixed, domain-specific model cast into hardware for a particular engineering purpose.

Early 19th century

Least Squares before Regression

Least squares was developed by Carl Friedrich Gauss for astronomical and geodetic problems.

Galton and “Regression to Mediocrity”

The concept originates with Francis Galton who studied the inheritance of human height.

He found that children of very tall parents are on average slightly shorter. Likewise, children of very short parents tend to be taller.

Formalization of Regression

Karl Pearson, Galton’s protegee, formalized regression and introduced the correlation coefficient.

1920s

Modern Statistical Framework

R. A. Fisher integrated regression into the emerging field or modern statistics by connecting:

- least squares
- maximum likelihood
- analysis of variance

Formalization into Multiple Regression

By 1950 regression expanded beyond simple two variable relationship.

Late 20th century

Expansion into Computational Regression

With computational power, regression evolved into generalized linear models, regularized regression, nonparametric regression and machine learning variants (e.g. gradient boosting, neural nets).

Before talking about ML, we should talk about data…

Recall the Advertising data from Module 1. Suppose that in our role as statistical consultants we are asked to suggest, based on given data, a marketing plan for next year that will result in high product sales. What information would be useful to provide such a recommendation?

Medium budget

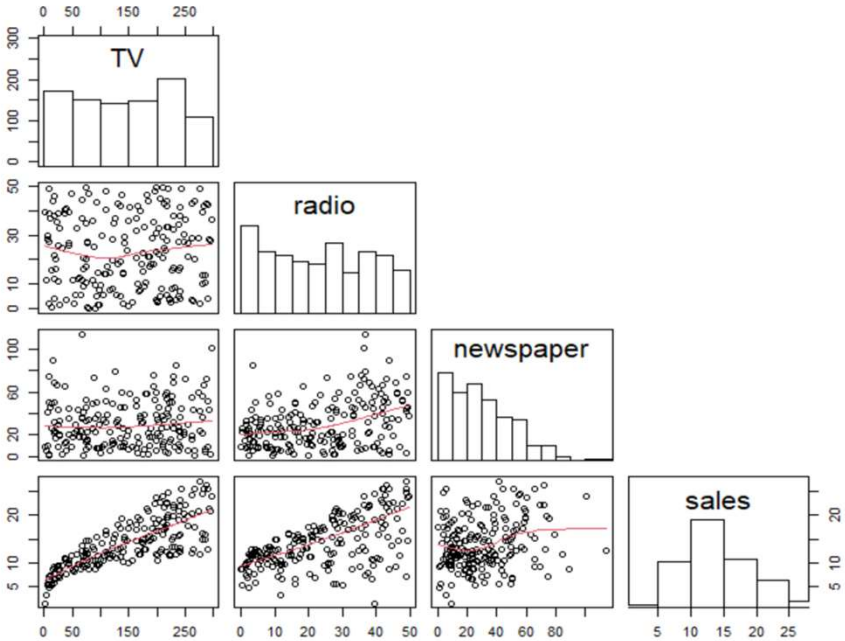
Sales

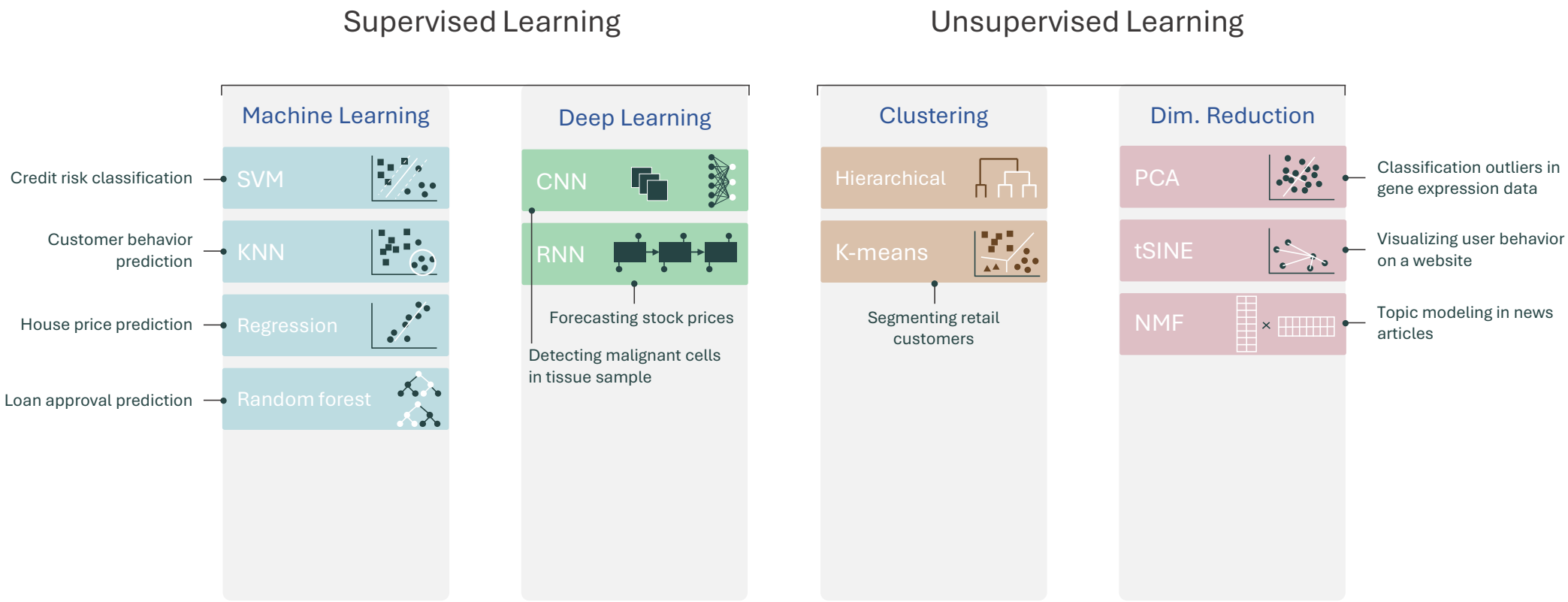
$$\hat{f}: \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

Markets
(market 1
market 2
...)

Predictor variable
(Feature)

Dependent variable
(Feature)



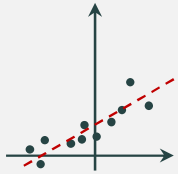

Supervised Learning
Unsupervised Learning
Machine Learning
Deep Learning
Clustering
Dim. Reduction
Credit risk classification
SVM
Customer behavior prediction
KNN
House price prediction
Regression
Loan approval prediction
Random forest
CNN
RNN
Forecasting stock prices
Detecting malignant cells in tissue sample
Hierarchical
K-means
Segmenting retail customers
PCA
Classification outliers in gene expression data
tSINE
Visualizing user behavior on a website
NMF
Topic modeling in news articles

Hypothesis and Learning Considerations:

1. Is there a relationship between advertising budget and sales?
2. How strong is the relationship between advertising budget and sales?
3. Which media are associated with sales?
4. How large is the association between each medium and sales?
5. How accurately can we predict future sales?
6. Is the relationship linear?
7. Is there synergy among the advertising media?

Linear Regression

1. Economic modeling
2. Marketing mix model
3. Customer life-time value



Continuous → Continuous

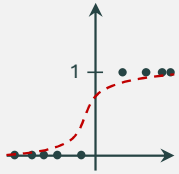
$$y = \beta_0 + \sum_{i=1}^{N} \beta_i x_i$$

lm(y ~ x1 + x2, data)

1 unit increase in x increases y by β

Logistic Regression

1. Customer choice model
2. Click-through rate
3. Conversion rate
4. Credit scoring



Continuous → Binary

$$y = \frac{1}{1 + e^{-z}}$$

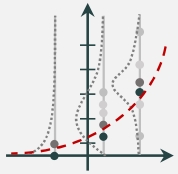
$$z = \beta_0 + \sum_{i=1}^{N} \beta_i x_i$$

glm(y ~ x1 + x2, data, family= binomial())

1 unit increase in x increases log odds by β

Poisson Regression

1. Number of orders in lifetime
2. Number of visits per user



Continuous → 0,1,2,...

$$y \sim \text{Poisson}(\lambda)$$

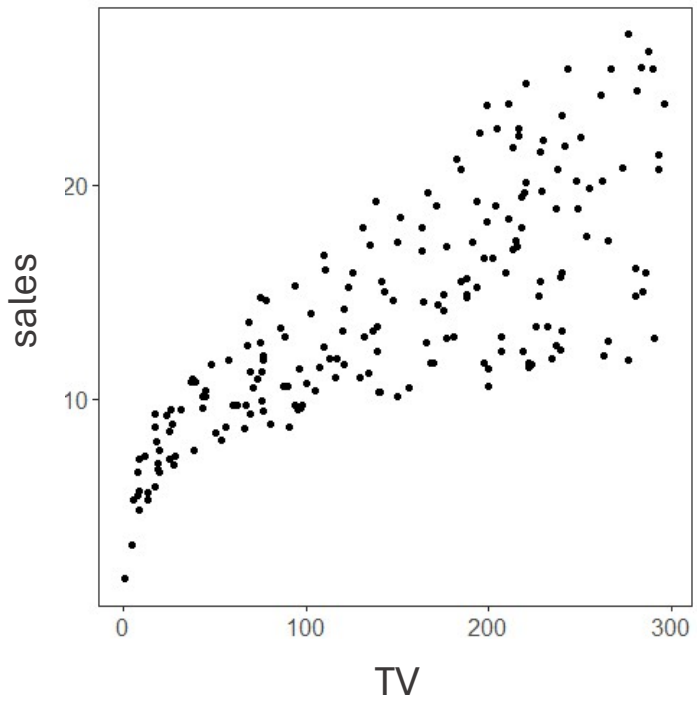
$$\ln \lambda = \beta_0 + \sum_{i=1}^{N} \beta_i x_i$$

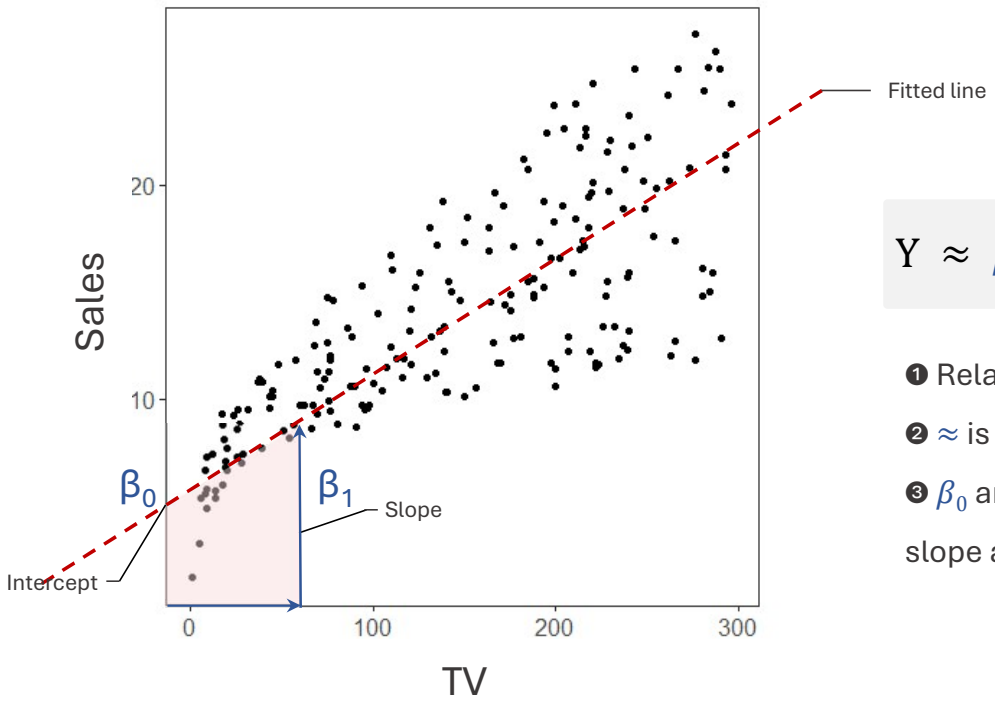
glm(y ~ x1 + x2, data, family= poisson())

1 unit increase in x increases multiplies y by e^{β}

❶ Is there a relationship between advertising budget and sales?

Our first goal should be to determine whether the data provide evidence of an association between advertising expenditure and sales. If the evidence is weak, then one might argue that no money should be spent on advertising!


sales
20
10
0
100
200
300
TV



$$Y \approx \beta_0 + \beta_1 X + \epsilon$$

❶ Relation between X and Y is assumed to be linear.

❷ $\approx$ is read *approximately modeled as...*

❸ β_0 and β_1 are unknown constants that represent slope and intercept.

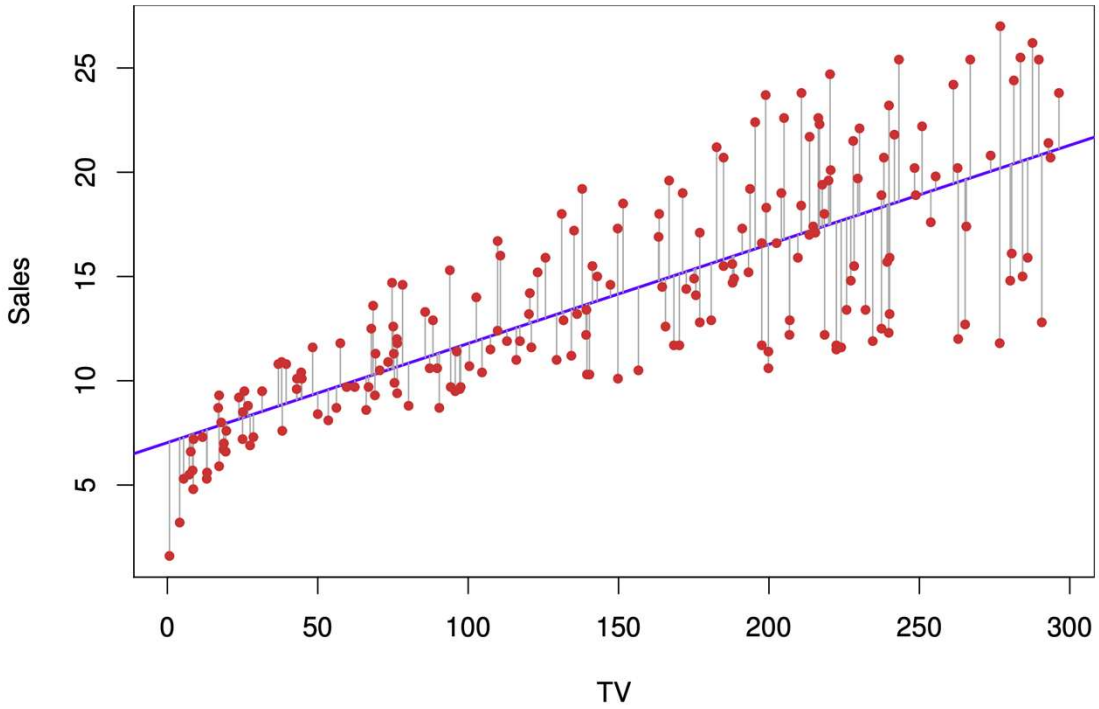
Estimating Coefficients

Our goal is to to find an intercept β_0 and a slope β_1 such that the resulting line is as close as possible to the $n = 200$ data points.

There are several ways of measuring closeness. However, by far the most common approach involves minimizing the least squares criterion.

Let $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_0 x_i$ be the prediction for Y based on the ith value of X. Then, $e_i = y_i - \hat{y}_i$ represents the ith residual – this is the difference between i^{th} observed and i^{th} predicted value.

We define the residual sum of squares as $RSS = e_1^2 + e_2^2 + e_n^2$. The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS.


Sales
5
10
15
20
25
0
50
100
150
200
250
300
TV

Dependent variable

Advertising data

Predictor variable

Model object

```
model <- data %$% lm(sales ~ TV)
summary(model)
```

```
##
## Call:
## lm(formula = sales ~ TV)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.3860 -1.9545 -0.1913  2.0671  7.2124
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 7.032594   0.457843   15.36   <2e-16 ***
## TV          0.047537   0.002691   17.67   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.259 on 198 degrees of freedom
## Multiple R-squared:  0.6119, Adjusted R-squared:  0.6099
## F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

Regression Coefficients β_0 β_1

Wrapper object for LM model

Wrapper object for LM engine

```
lm_model <- linear_reg() %>% set_engine("lm")
lm_form_fit <- lm_model %>% fit(sales ~ TV, data = data)
summary(lm_form_fit$fit)
```

```
##         Length Class      Mode
## lvl      0     -none-     NULL
## spec     7     linear_reg list
## fit     12     lm         list
## preproc  1     -none-     list
## elapsed  1     -none-     list
```

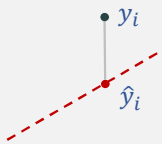
Model object

When finding linear regression coefficients breaks down...

Ordinary least squares normally gives coefficients through a precise linear-algebra solution, so no iterative search is required. But in real-world settings with massive datasets, streaming data, or strong regularization, computing the exact solution becomes impractical or unstable. In these niches, iterative methods like Gradient Descent or Coordinate Descent become essential because they scale better, handle constraints, and adapt to data that cannot be processed in one shot.

Loss Functions

❶ Error measure for a single training example
❷ Quantify how wrong the model is on one observation



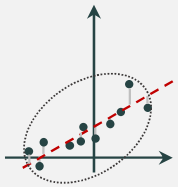
Notation: $L_i(\theta)$

Squared Error (L2)
$(y_i - \hat{y}_i)^2$
Absolute Error (L1)
$|y_i - \hat{y}_i|^2$
Huber Loss (Both)
$\frac{1}{2}\left((y_i - \hat{y}_i)^2\right)$ if $|y_i - \hat{y}_i|^2 < \delta$
$\delta|y_i - \hat{y}_i| - \frac{1}{2}\delta^2$ otherwise

Input: one pair (x_i, y_i)
Output: a single scalar per sample

Cost Functions

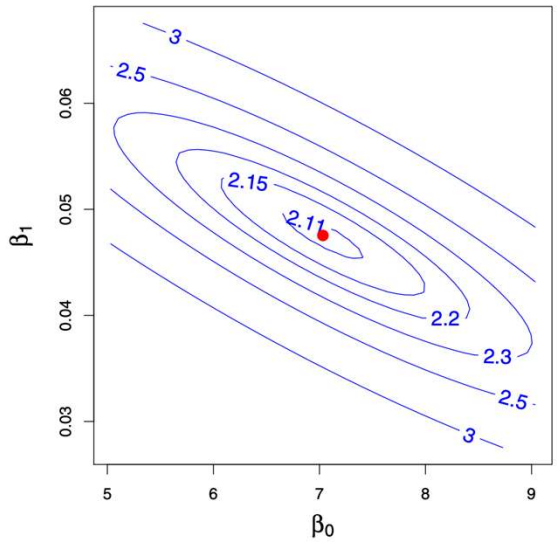
❶ Aggregated error measure for the entire dataset
❷ Provide a global objective to optimize model parameters

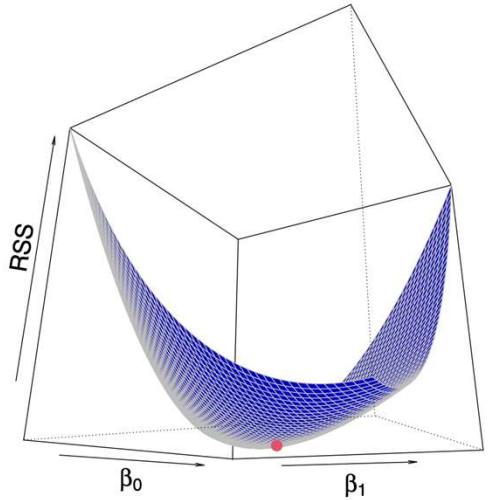


Notation: $J(\theta)$

RSS, AE,
Mean Squared Error
$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$
MAE

Input: all pairs $\{(x_i, y_i)\}_{i=1}^{n}$
Output: a single scalar summarizing the model's performance





1. Linear Regression (when solved iteratively)

Ordinary LR has a closed-form solution, but gradient descent can be use when:

❶ The dataset is huge
❷ The matrix is too large to invert
❸ You are using stochastic or mini-batch training

Use Case:
Common in large-scale ML systems

2. Logistic Regression and other GLMs

Logistic regression has no closed-form solution (the sigmoid makes the likelihood nonlinear) and thus is typically optimized with:

❶ Gradient Descent or Stochastic Gradient Descent
❷ Newton-type methods

Use Case:
Extremely fast for large-scale e-commerce datasets that is easy to distribute across machines or run on GPUs

3. Support Vector Machines

Classical SVM use quadratic programming, but linear SVMs (e.g. in large-scale libraries) often use:

❶ Stochastic Gradient Descent
❷ Pegasos (a stochastic gradient method)
❸ Hinge-loss Gradient Descent

Use Case:
Large-scale text classification systems such as spam filters

4. Clustering Algorithms

Some clustering algorithms can be optimized with Gradient Descent:

❶ K-means variants (e.g. soft k-means)
❷ Gaussian mixture models (e.g. gradient-based EM alternatives)
❸ Spectral clustering approximation

Use Case:
Image compression, audio modeling and community detection in massive graphs

5. Dimensionality Reduction

Dimensionality reduction often involves optimizing an objective function that measures how well a low-dimensional representation preserves structure in the data. Many of these objectives are nonlinear, non-convex, huge and impossible to solve with closed-form linear algebra:

❶ Principal Component Analysis for huge datasets
❷ t-SNE and UMAP

Use Case:
Coordinate Search doesn't care. It just tries directions and picks the best.

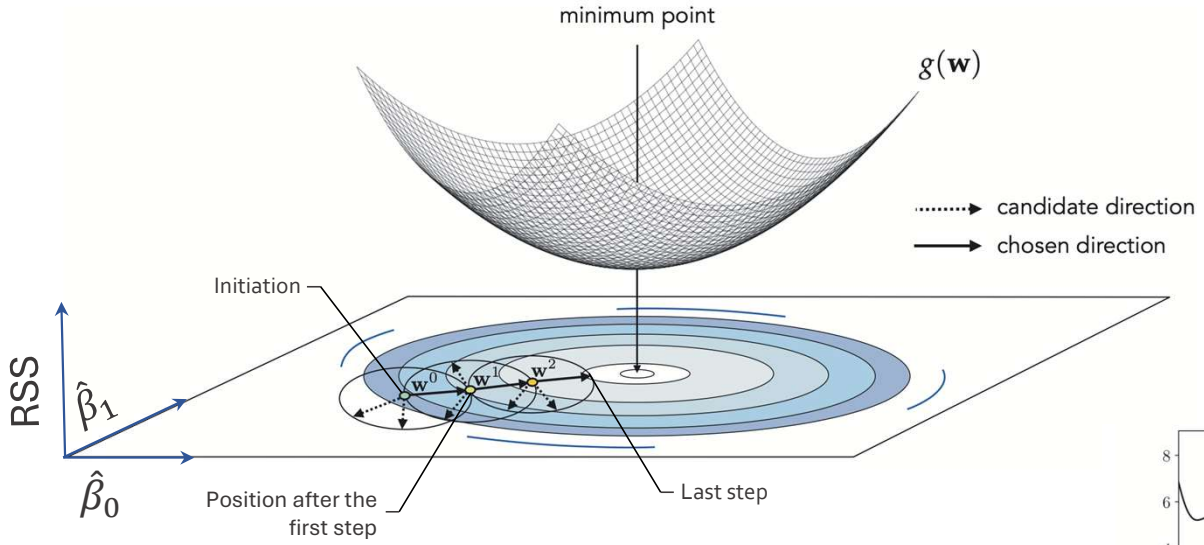
Gradient Descent

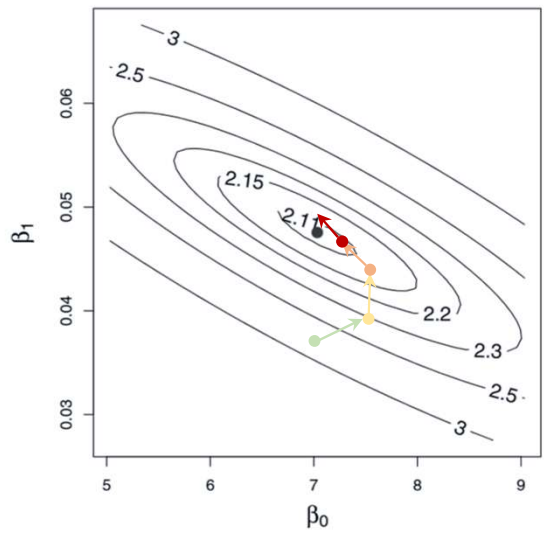
Suppose $J(\theta)$ is the loss function and θ is the parameters that the ML model will learn. Suppose we are making the loss function continuous functions, and the model wants to learn those parameters for which the cost will be minimal.

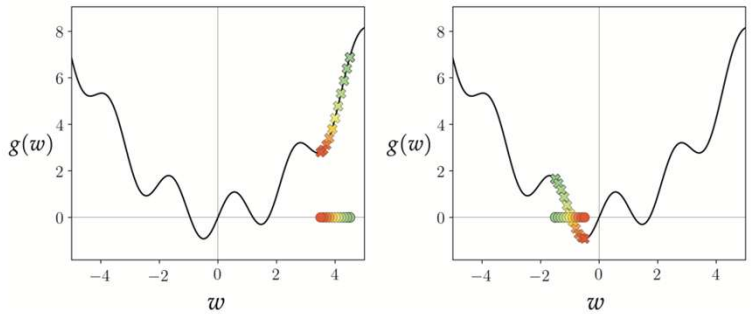
Run multiple times.

> Initialize the parameter (weight) values randomly.
>
> > Partially differentiate the cost function $G = {}^{\delta J(\theta)}/_{\delta\theta}$ with respect to different parameters constituting the cost function.
> >
> > Update the parameters (weights) by an amount proportional to the gradient to ensure that the loss reduces in each iteration, $\theta = \theta - \alpha G$. Here, α is the learning rate parameter which is considered a vital hyperparameter in deciding how the fast the parameters' updates should happen.
>
> Repeat the process of updation until the difference in values of the cost function between two consecutive iterations goes below the pre-defined threshold.

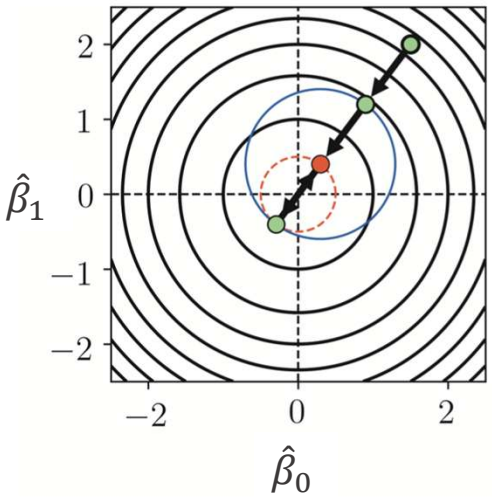
Return model parameters (weights) for which G value is the lowest.


minimum point
$g(\mathbf{w})$
candidate direction
chosen direction
Initiation
RSS
$\hat{\beta}_1$
$\hat{\beta}_0$
$\mathbf{w}^0$
$\mathbf{w}^1$
$\mathbf{w}^2$
Position after the first step
Last step

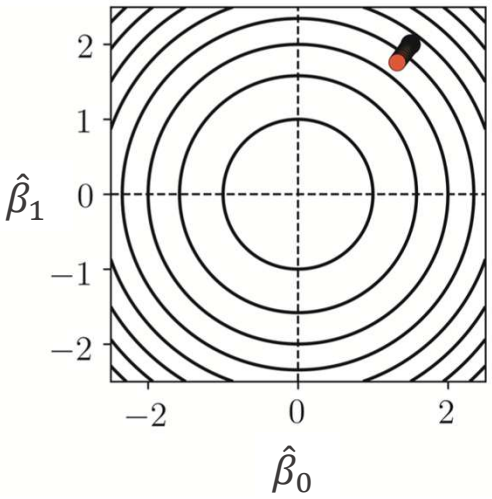

3
2.5
2.15
2.11
2.2
2.3
2.5
3
0.06
0.05
0.04
0.03
β_1
5
6
7
8
9
β_0


$g(w)$
w
$g(w)$
w

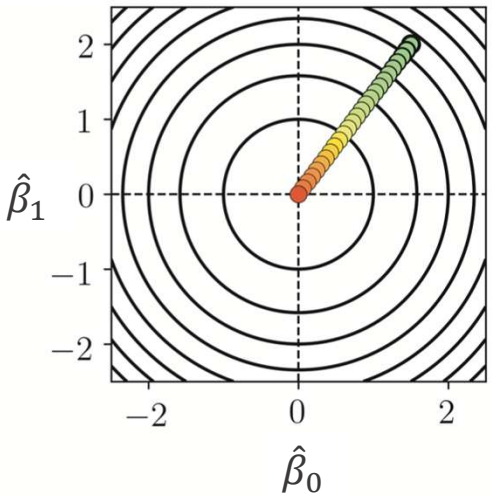
Stepping parameter α



Small: Smaller learning rate requires many updates; hence learning would be slow.



Large: Larger learning rate may lead to drastic updates and GD may diverge from the minima.



Optimal learning rate smoothly reaches to the minima.

1.
Sometimes you cannot compute a gradient

Examples:

❶ The function is noisy
❷ The function is not differentiable
❸ The function is a black box (e.g. simulator)
❹ The function is expensive to evaluate
❺ The function has discontinuities

Consequence:
Gradient descent simply cannot run

Solution:
Coordinate Search survives here because it doesn't need derivatives. It just tries "move left or right" along each axis.

2.
Gradients are too expensive to compute

Examples:

❶ Models with billions of parameters
❷ A cost function that requires huge computation per gradient

Consequence:
Computing the full gradient vector might be too slow

Solution:
Coordinate Descent updates one parameter at a time, which can be much cheaper.

3.
Problems is 'one-coordinate-at-a-time'

Certain optimization problems have a structure where each variable can be optimized independently or cheaply.

Examples:

❶ Lasso regression, Elastic net
❷ Sparse models, problems with separable penalties

Consequence:
Gradient Descent does not exploit this structure

Solution:
Coordinate Descent shines here because each update is simple, fast and often closed-form.

4.
Many classical ML problems are sparse

If most variables are zero or irrelevant...

Consequence:

❶ Gradient Descent do a lot of unnecessary work
❷ Update only all coordinates instead of meaningful ones
❸ Converge slower than alternative solutions

Solution:
Coordinate methods are faster, update only meaningful coordinates and converge faster than Gradient Descent.

5.
Gradient Descent struggles when the landscape is ugly

Examples:

❶ The surface is jagged
❷ The function is not smooth
❸ The gradient is unreliable
❹ The Loss function has plateau or cliffs

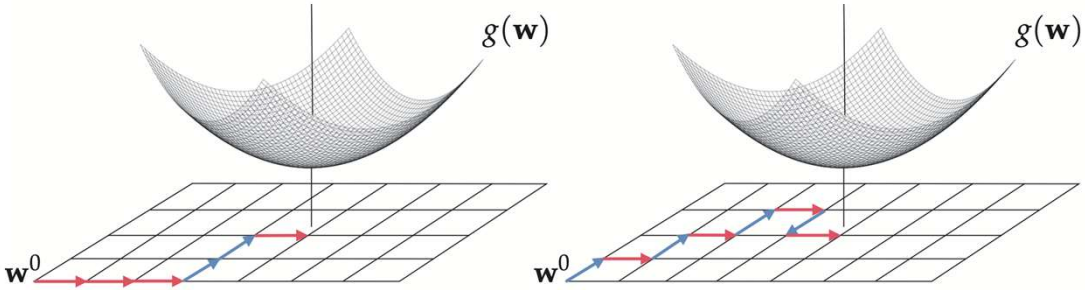
Solution:
Coordinate Search doesn't care. It just tries directions and picks the best.

Coordinate Search/Descent

Initialize θ and repeat until convergence.

> Optimize one coordinate at a time.
>
> > Coordinate search (heuristic method): try a small forward and backward step while keeping all other parameters fixed.
> >
> > Coordinate descent (true optimization method using the gradient or a closed-form optimizer): Hold all other coordinates fixed and minimize the objective with respect to the ith coordinate $G_i = \partial J(\theta) / \partial\theta[i]$. Update only this coordinate $\theta[i] = \theta[i] - \alpha * G_i$.

Return model parameters for which cost is the lowest.



❷ How strong is the relationship between advertising budget and sales?

Assuming that there is a relationship between the advertising budget and sales, we would like to know how strong is the relationship. Does knowledge of the advertising budget provide a lot of information about product sales?

Dependent variable

Advertising data

Predictor variable

```
model <- data %$% lm(sales ~ TV)
summary(model)
```

```
##
## Call:
## lm(formula = sales ~ TV)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
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## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.259 on 198 degrees of freedom
## Multiple R-squared:  0.6119, Adjusted R-squared:  0.6099
## F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```
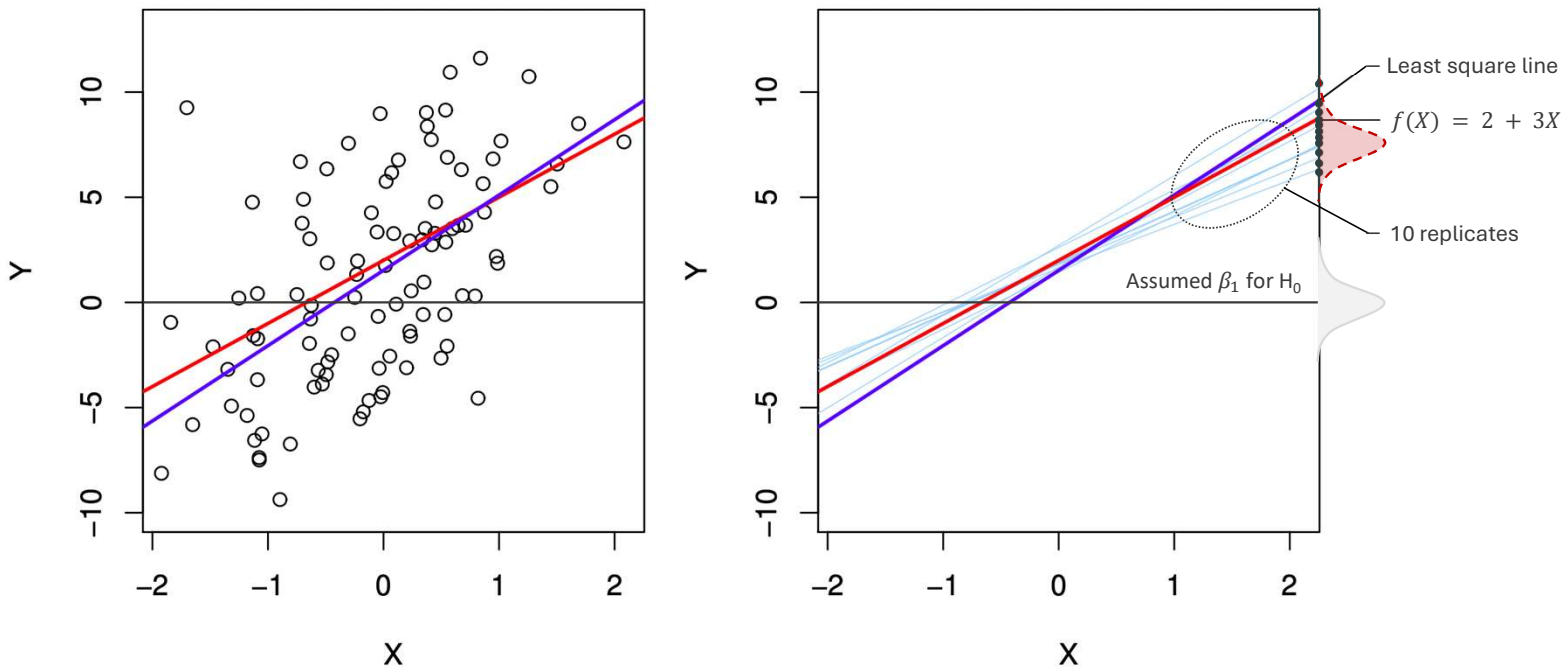

Regression Coefficients: β_0, β_1

Hypothesis Testing

H_0: There is no relationship between $X(TV)$ and $Y(Sales)$ ⟶ **H_0:** $\beta_1 = 0$

H_1: There is some relationship between $X(TV)$ and $Y(Sales)$ ⟶ **H_1:** $\beta_1 \neq 0$


Y
X
Least square line
$f(X) = 2 + 3X$
10 replicates
Assumed β_1 for H_0

H_0: There is no relationship between $X(TV)$ and $Y(Sales)$ $\longrightarrow$ **H_0:** $\beta_1 = 0$

H_1: There is some relationship between $X(TV)$ and $Y(Sales)$ $\longrightarrow$ **H_1:** $\beta_1 \neq 0$

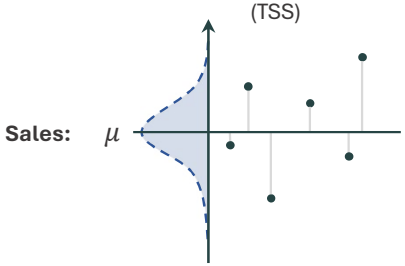
	Variance	Standard Error	95% Confidence Interval	t-statistics	p-Value
What it Measures	Variability of the data around the mean	Uncertainty of each coefficient estimate	Range of plausible values for the true coefficient	How many Ses the estimate is away from 0	Probability of observing such a t-value if the true coefficient were 0
Formula	$Var(\hat{\mu}) = SE(\hat{\mu})^2 = \frac{\sigma^2}{n}$	$SE(\hat{\beta}_0)^2 = \sigma^2\left[\frac{1}{n} + \frac{x^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right]$ $SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$	$\hat{\beta}_0 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_0)$ $\hat{\beta}_1 \pm 2 \cdot \mathrm{SE}(\hat{\beta}_1)$	$t = \frac{\hat{\beta}_1}{\mathrm{SE}(\hat{\beta}_1)}$	
What Appears in *lm()* Output	Not shown directly; used internally	Column **Std. Error**	Not in default output; shown via *confint(model)*	**Column *t* value**	Column $\boldsymbol{P}(> \lvert \boldsymbol{t} \rvert)$
How it Explains the Readout	Drives the Residual standard error, SEs, and overall model fit. High variance → noisier residuals	Tells you how precisely each coefficient is estimated. Small SE → stable, reliable coefficient	Helps interpret whether the effect could be zero. If CI excludes 0 → likely meaningful effect	Measures strength of evidence. Large t → coefficient β_1 is far from zero relative to noise	Determines statistical significance. Small p → strong evidence the coefficient β_1 is not zero

Variance

```
## Residual standard error: 3.259 on 198 degrees of freedom
## Multiple R-squared:  0.6119, Adjusted R-squared:  0.6099
## F-statistic: 312.1 on 1 and 198 DF,  p-value: < 2.2e-16
```

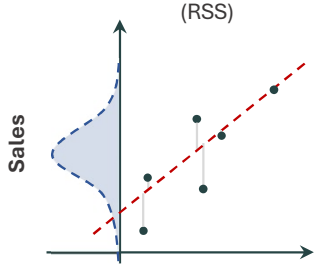
```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.3860 -1.9545 -0.1913  2.0671  7.2124
```

Total Sum of Squares (TSS)



Existing variance in **Sales** dependent variable

Residual Sum of Squares (RSS)



Variance in **Sales** explained by **TV**

$$R^2 = 1 - \frac{RSS}{TSS}$$

61%

of the total variability in **Sales** captured by the predictor

0 → model explains none of the variability
1 → model explains all the variability

$$RSE = \sqrt{\frac{1}{n-p} RSS}$$

With a **Sales** mean of 14.022 and a **RSE** of 3.259 units, the model's typical prediction is

23%

off.

❸ Which media are associated with sales?

We want to know whether all three media — TV, radio, and newspaper — are related to sales, or whether only one or two of them truly matter. To answer this, we need a way to isolate the unique contribution of each medium, given that spending occurred across all three.

Simple regression of Sales on TV

	Estimate	Std. error	*t*-statistics	*p*-value	RSE	Multiple R^2
Intercept	7.0325	0.4578	15.36	<0.0001	3.26 (23%)	0.61
TV	0.0475	0.027	17.67	<0.0001		

Simple regression of Sales on Radio

	Estimate	Std. error	*t*-statistics	*p*-value	RSE	Multiple R^2
Intercept	9.312	0.536	16.54	<0.0001	4.27 (30%)	0.33
Radio	0.203	0.020	9.92	<0.0001		

Simple regression of Sales on Newspaper

	Estimate	Std. error	*t*-statistics	*p*-value	RSE	Multiple R^2
Intercept	12.251	0.621	19.88	<0.0001	5.09 (36%)	0.05
Newspaper	0.055	0.017	3.30	0.00115		

What Happens in when Separate Regression Models are Fitted?

Omitted-variable bias	Each model ignores other predictors, so coefficients become biased if predictors are correlated.
Inflated Significance	Predictors may look significant only because they proxy for other variables.
Inconsistent Interpretation	Each model answers a different question ("effect of X alone"), making results incomparable.
Model selection confusion	Hard to know which simple model is "best"; results conflict.
Overestimation of R^2	Each simple model may appear to explain a lot, but only because variance is not partitioned correctly.
No insight into multicollinearity	Correlations among predictors remain invisible.

Dependent variable

Advertising data

Predictor 1

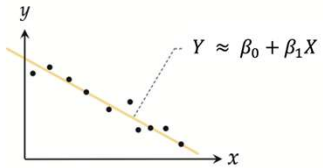
Predictor 2

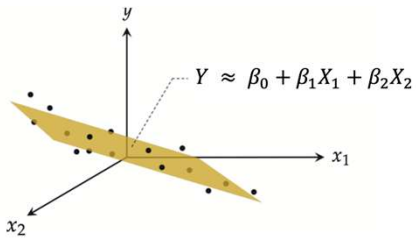
Predictor 3

```
model <- data %$% lm(sales ~ TV + radio + newspaper)
summary(model)
```

```
##
## Call:
## lm(formula = sales ~ TV + radio + newspaper)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.8277 -0.8908  0.2418  1.1893  2.8292
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.938889   0.311908   9.422   <2e-16 ***
## TV           0.045765   0.001395  32.809   <2e-16 ***
## radio        0.188530   0.008611  21.893   <2e-16 ***
## newspaper   -0.001037   0.005871  -0.177     0.86
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.686 on 196 degrees of freedom
## Multiple R-squared:  0.8972, Adjusted R-squared:  0.8956
## F-statistic: 570.3 on 3 and 196 DF,  p-value: < 2.2e-16
```

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$$





	What Happens in when Separate Regression Models are Fitted?	Why Multiple Regression Avoids It
Omitted-variable bias	Each model ignores other predictors, so coefficients become biased if predictors are correlated.	Includes all predictors simultaneously, removing bias from omitted variables.
Inflated Significance	Predictors may look significant only because they proxy for other variables.	Tests each predictor controlling for the others, giving correct significance.
Inconsistent Interpretation	Each model answers a different question ("effect of X alone"), making results incomparable.	Each model answers a different question ("effect of X alone"), making results incomparable.
Model selection confusion	Hard to know which simple model is "best"; results conflict.	A single model provides a unified assessment of all predictors.
Overestimation of R^2	Each simple model may appear to explain a lot, but only because variance is not partitioned correctly.	Multiple regression partitions shared variance properly.
No insight into multicollinearity	Correlations among predictors remain invisible.	Multiple regression reveals collinearity through SE inflation and diagnostics.

Simple regression of Sales on TV

	Estimate	Std. error	*t*-statistics	*p*-value	RSE	Multiple R^2
Intercept	7.0325	0.4578	15.36	<0.0001	3.26 (23%)	0.61
TV	0.0475	0.027	17.67	<0.0001		

Simple regression of Sales on Radio

	Estimate	Std. error	*t*-statistics	*p*-value	RSE	Multiple R^2
Intercept	9.312	0.536	16.54	<0.0001	4.27 (30%)	0.33
Radio	0.203	0.020	9.92	<0.0001		

Simple regression of Sales on Newspaper

	Estimate	Std. error	*t*-statistics	*p*-value	RSE	Multiple R^2
Intercept	12.251	0.621	19.88	<0.0001	5.09 (36%)	0.05
Newspaper	0.055	0.017	3.30	0.00115		

Simple regression of Sales on Newspaper

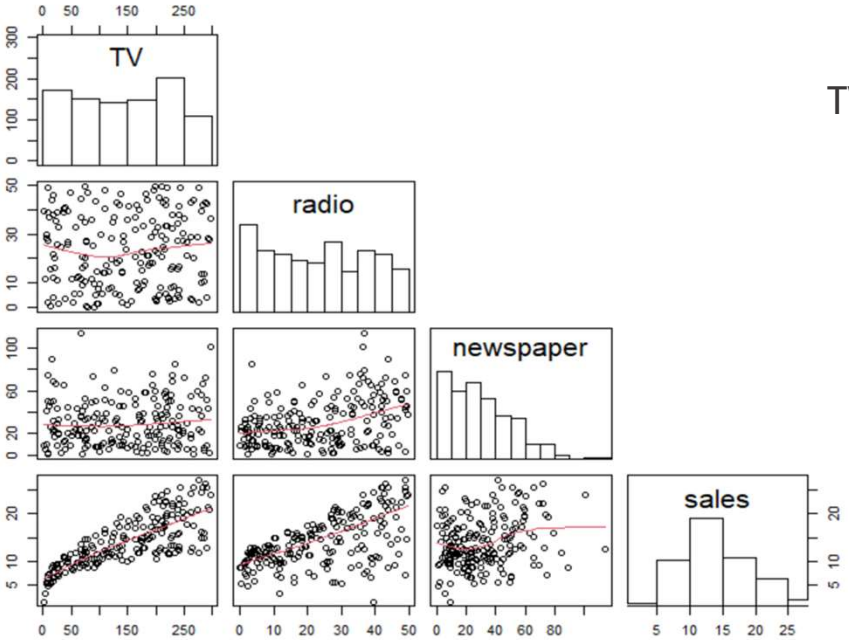
	Estimate	Std. error	*t*-statistics	*p*-value
Intercept	12.251	0.621	19.88	<0.0001
Newspaper	0.055	0.017	3.30	0.00115

RSE	Multiple R^2
5.09 (36%)	0.05

Multiple regression of Sales on TV+ Radio + Newspaper

	Estimate	Std. error	*t*-statistics	*p*-value
Intercept	2.938	0.3119	9.422	<0.0001
TV	0.045	0.0014	32.809	0.00115
Radio	0.188	0.0086	21.893	<0.0001
Newspaper	-0.001	0.0058	-0.177	0.86

RSE	Multiple R^2
1.686 (12%)	0.8972



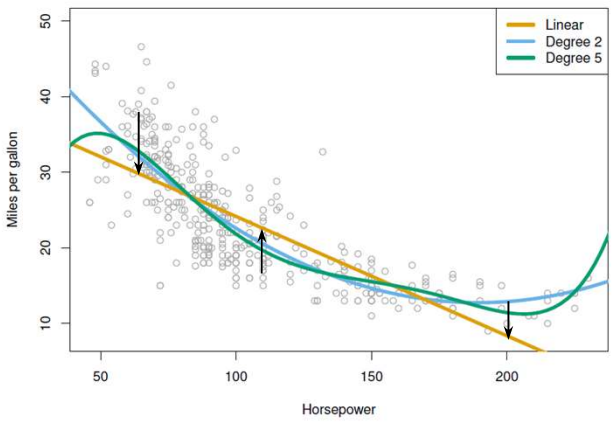
TV	0.0548	0.0567	0.7822
Radio		0.3541	0.5762
Newspaper			0.2283
			Sales

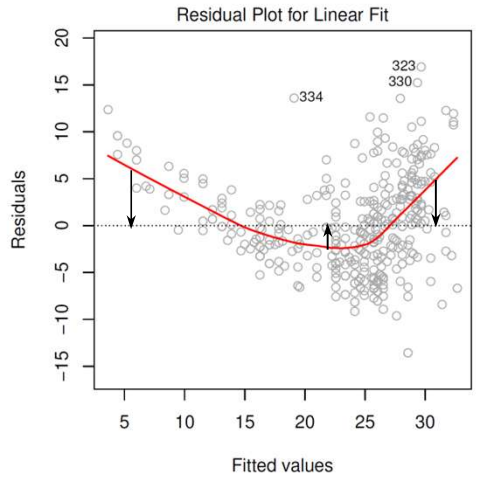
	Collinearity	Additive Effect
What it describes	Relationship between predictors (e.g., Newspaper and Radio move together)	Relationship between predictors and the response (e.g., TV and Radio jointly influence Sales)
Mechanism	Predictors share overlapping information → hard to separate their individual contributions	Predictors amplify or modify each other's effect on Sales
Effect on Coefficients	Inflated standard errors, unstable estimates, misleading p-values	Additional term (e.g., TV × Radio) captures synergy or non-additivity
Interpretation Challenge	Hard to tell which predictor "deserves" credit	Shows that the combined effect is more (or less) than the sum of parts
Case	Newspaper and Radio are moderately correlated → Newspaper becomes insignificant in multiple regression	TV and Radio interaction improves model fit → marketing synergy
Modeling implication	Often requires regularization or variable selection	Requires including interaction terms to capture real structure
Goal	Reduce or diagnose	Understand and model

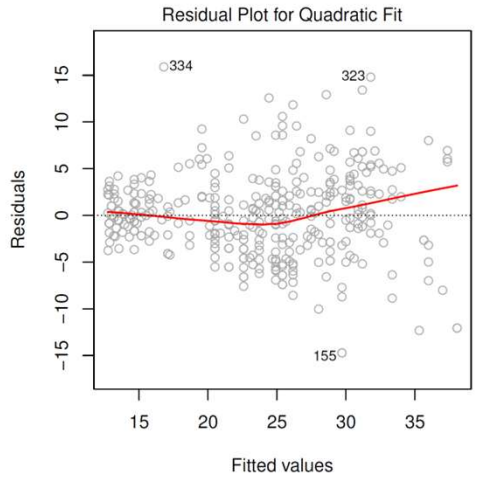
Potential Problems:

1. Non-linearity of the response-predictor relationships.
2. Correlation of the error terms.
3. Non-constant variance of error terms.
4. Outliers.
5. High-leverage points.
6. Collinearity.

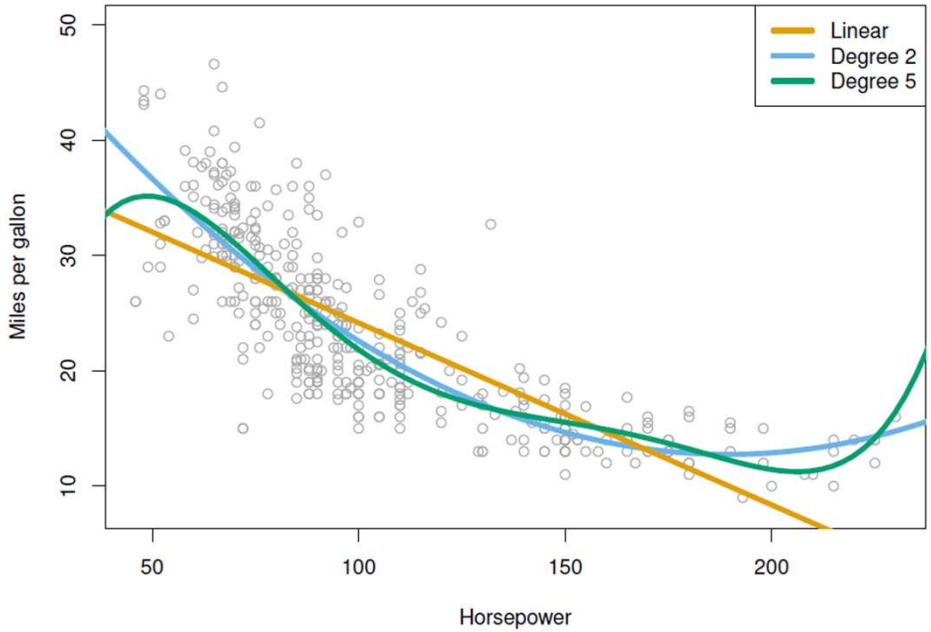
Non-Linearity of the Response-Predictor Relationships







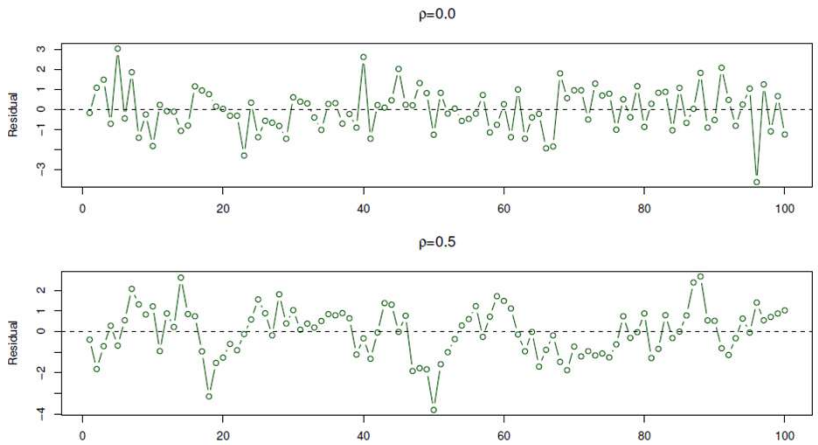
Assumption Violated	Effect	Diagnostic Tools to Spot This	How to Fix This	Cost of Fixing
The linear regression model assumes that there is a straight-line relationship between the predictors and the response.	If true relationship is far from linear, then virtually all conclusions we draw from the fit are suspect.	Residual plot diagnostics: plot the residuals $\epsilon_i = y_i - \hat{y}_i$ versus the predictor $\hat{x}_i$. In the case of multiple regression model, we plot the residuals versus the predicted values $\hat{y}_i$.	Use $logX, \sqrt{X}, and\ X^2$.	**Loss of explainability** since the original units of the predictor has been transformed.

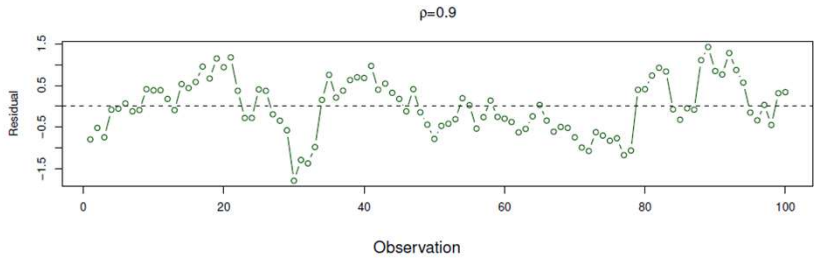


$$mpg = \beta_0 + \beta_1 \times horsepower + \beta_2 \times horsepower^2$$

	Estimate	Std. error	*t*-statistics	*p*-value
Intercept	56.9001	1.8004	31.6	<0.0001
horsepower	-0.4662	0.0311	-15.0	<0.0001
horsepower²	0.0012	0.0001	10.1	<0.0001

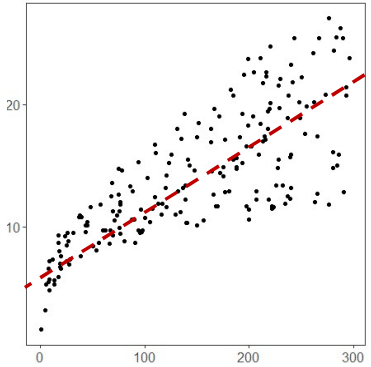
Correlation of the Error Terms

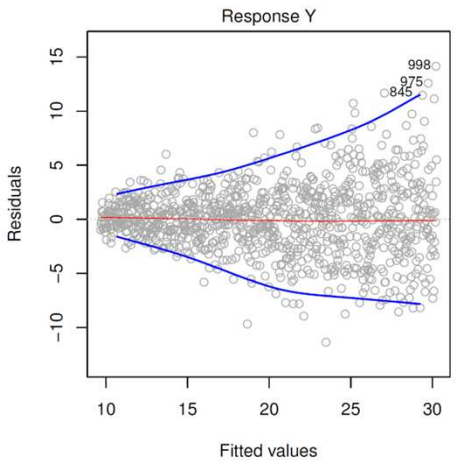


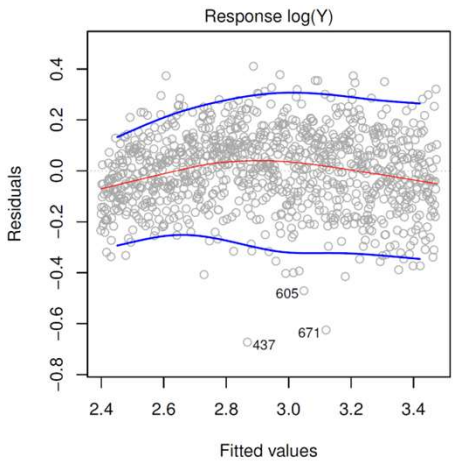


Assumption Violated	Effect	Diagnostic Tools to Spot This	How to Fix This	Cost of Fixing
The linear regression model assumes that error terms $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$ are uncorrelated. This means that if, e.g. ϵ_1 is positive, this provides no information about $\epsilon_1 + 1$.	The standard errors that are computed for the regression coefficients or fitted values assume the uncorrelated error terms. As a result, **confidence intervals** will be narrower that they should. **P-values** will be lower than they should be.	Residual plot diagnostics: plot the residuals $\epsilon_i = y_i - \hat{y}_i$ for each observation. Sort by increasing index.	Change your experimental design: do not use time series in a simple/multiple linear regression model.	

Non-Constant Variance of Error Terms

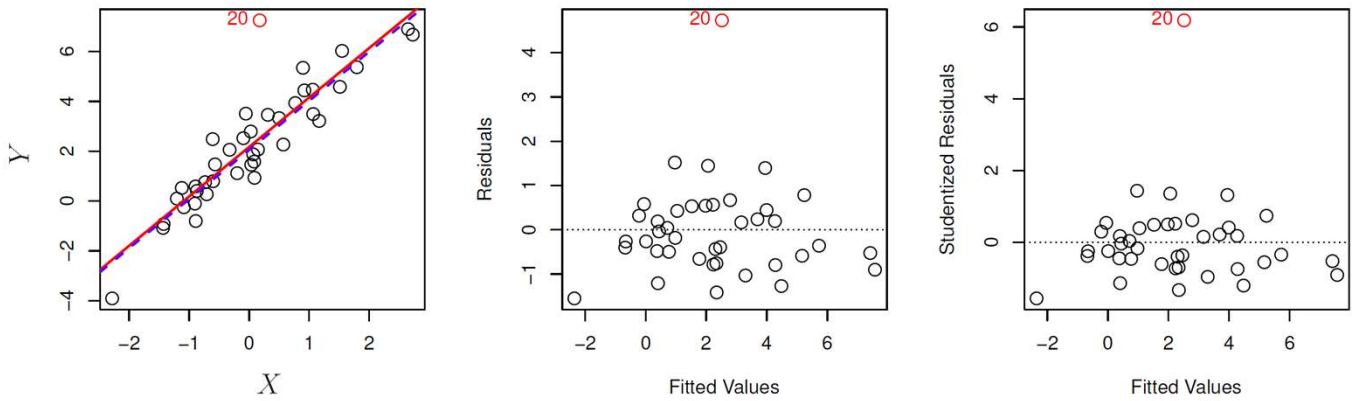






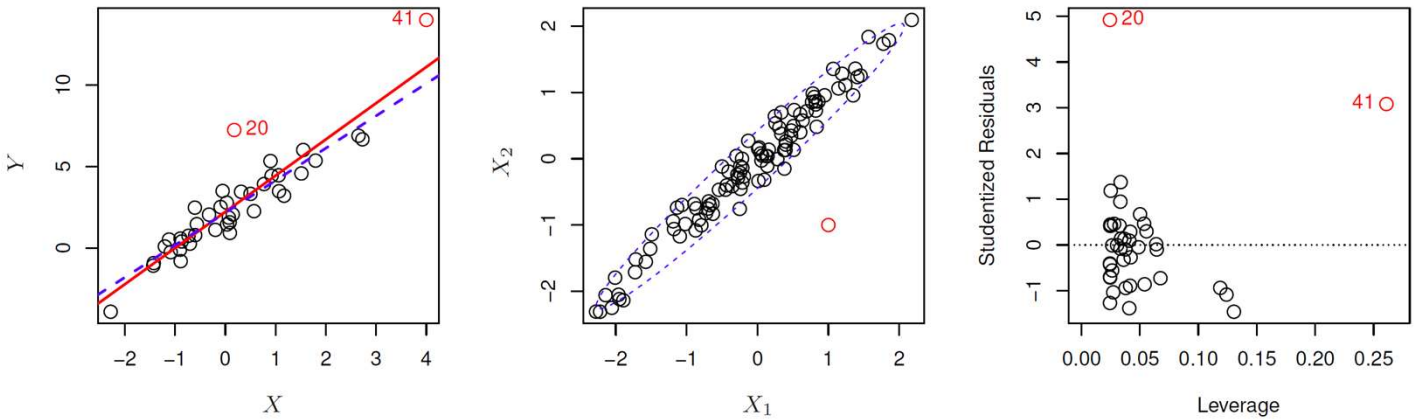
Assumption Violated	Effect	Diagnostic Tools to Spot This	How to Fix This	Cost of Fixing
Homoscedasticity: the linear regression model assumes that error terms have constant variance $Var(\epsilon_i) = \sigma^2$.	The estimation of standard errors, confidence intervals and hypothesis testing rely on this assumption.	Residual plot diagnostics: plot the residuals $\epsilon_i = y_i - \hat{y}_i$ versus the predictor $\hat{x}_i$. In the case of multiple regression model, we plot the residuals versus the predicted values $\hat{y}_i$.	Transform with concave function such as $logY$ or $\sqrt{Y}$. Or use methods such as weighted least squares.	**Loss of explainability** since the original units of the dependent value has been transformed.

Outliers



Assumption Violated	Effect	Diagnostic Tools to Spot This	How to Fix This	Cost of Fixing
An outlier is a point that is too far from the predicted value by the model.	Little effect on least squares fit coefficients. Profound effect on RSE (1.09 with pt 20, 0.77 without it) → confidence intervals and p-value. Similarly, inclusion of a pt. 20 causes R^2 to decrease from 0.892 to 0.805.	Residual plot diagnostics suffice in most cases with simple linear regression. Studentized residuals (each ϵ_i is divided by estimated SE) provide more advanced way to decide on whether a point should be considered residual or not.	Remove from training observations used for model fitting.	Need to retrain the model.

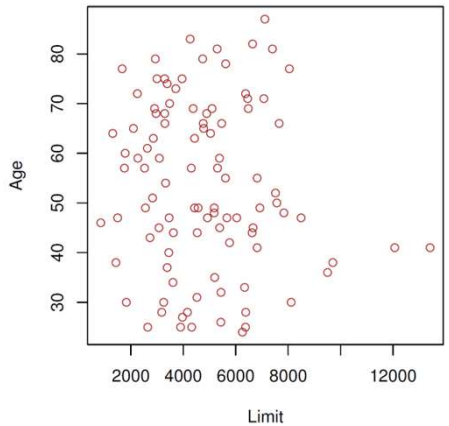
High Leverage Points



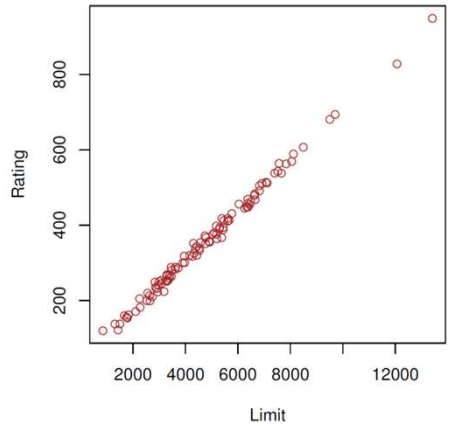
Assumption Violated	Effect	Diagnostic Tools to Spot This	How to Fix This	Cost of Fixing
Contrary to outliers, high leverage points have very unusual value of predictor x.	Substantial impact on least squares fit coefficients.	Use studentized residuals vs. leverage statistics h_i (always between values of $1/n$ and 1) in multiple-regression cases with more than 2 predictors. High leverage points are those with leverage statistics exceeding $(p+1)/n$.	Remove from training observations used for model fitting.	Need to retrain the model.

Multi-Collinearity of Predictors

Balance ~ Age + Limit

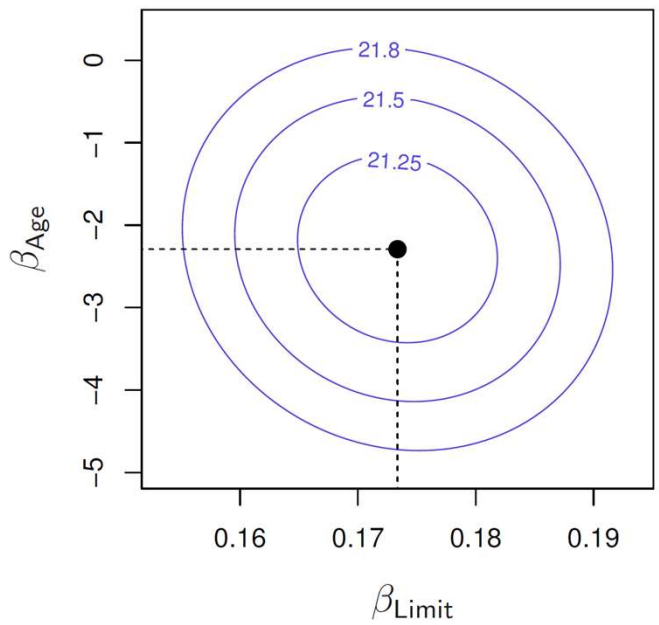


Balance ~ Rating + Limit

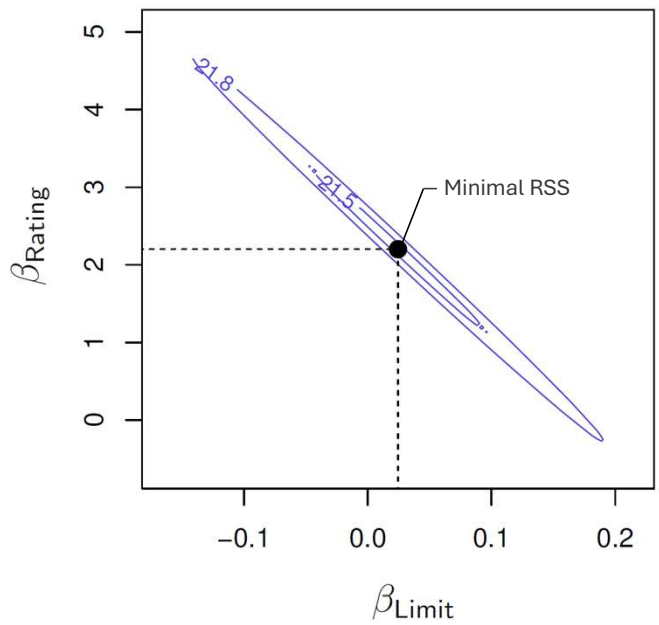


Assumption Violated	Effect	Diagnostic Tools to Spot This	How to Fix This	Cost of Fixing
No predictor is an exact linear combination of other predictors.	Substantial impact on the accuracy of the regression coefficients, causing the standard error to grow. Consequently, collinearity results in decline of t-statistics.	In some cases, use correlation matrix of the predictors. Alternative way is to calculate variance inflation factor, VIF. $$VIF_j = \frac{1}{1 - R^2{}_j}$$ Common rule-of-thumb thresholds: • > 5 → moderate collinearity • > 10 → severe collinearity	Drop some of the problematic variables.	Need to retrain the model.

Balance ~ Age + Limit



Balance ~ Rating + Limit



Multiple regression of Balance on Age and Limit

	Estimate	Std. error	*t*-statistics	*p*-value
Intercept	-173.411	43.828	-3.957	<0.0001
Age	-2.292	0.672	-3.407	0.0007
Limit	0.173	0.005	34.496	<0.0001

Multiple regression of Balance on Rating and Limit

	Estimate	Std. error	*t*-statistics	*p*-value
Intercept	-377.537	45.254	-8.343	<0.0001
Rating	2.202	0.952	2.312	0.0213
Limit	0.025	0.064	0.384	0.7012

Balance ~	Age +	Rating +	Limit	
	1.01	160.67	160.59	**VIF**

	Limit +	Limit -	
	0.754	0.750	**R^2**

The Marketing Plan

❶ *Is there a relationship between sales and advertising budget?*

This question can be answered by fitting a multiple regression model of Sales onto TV, Radio, and Newspaper, and testing the hypothesis $H_0\colon \beta_{TV} = \beta_{Radio} = \beta_{Newspaper} = 0$. We showed that associated p-value to the F-statistic can be used to determine whether or not we should reject this null hypothesis, and this one is indeed very low.

❷ *How strong is the relationship?*

We discussed two measures of model accuracy. First, the RSE estimates the standard deviation of the response from the population regression line. For the Advertising data, the RSE is 1.69 units while the mean value for the response is 14.022, indicating a percentage error of roughly 12 %. Second, the R^2 statistic records the percentage of variability in the response that is explained by the predictors. The predictors explain almost 90 % of the variance in sales.

The Marketing Plan

❸ *Which media are associated with sales?*

To answer this question, we can examine the p-values associated with each predictor's t-statistic. In the multiple linear regression, the p-values for TV and Radio are low, but the p-value for Newspaper is not. This suggests that only TV and radio are related to sales.

TV	0.0548	0.0567	0.7822
	Radio	0.3541	0.5762
		Newspaper	0.2283
			Sales

	Estimate	Std. error	*t*-statistics	*p*-value
Intercept	2.938	0.3119	9.422	<0.0001
TV	0.045	0.0014	32.809	0.00115
Radio	0.188	0.0086	21.893	<0.0001
Newspaper	-0.001	0.0058	-0.177	0.86

The Marketing Plan

❹ *How large is the association between each medium and sales?*

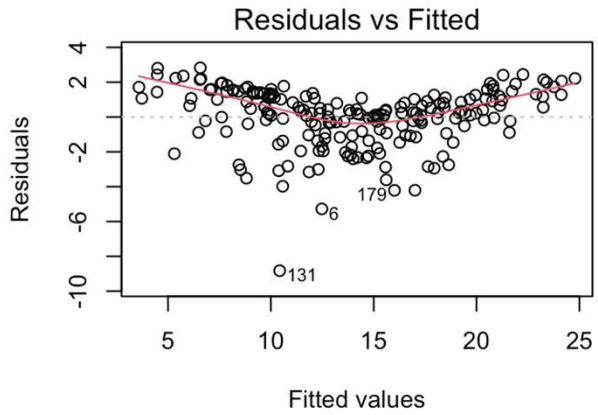
The standard error of β_j can be used to construct **confidence intervals** for β_j. For the Advertising data, we can use compute the 95 % confidence intervals for the coefficients in a multiple regression model using all three media budgets as predictors. The confidence intervals for TV and radio are narrow and far from zero, providing evidence that these media are related to sales.

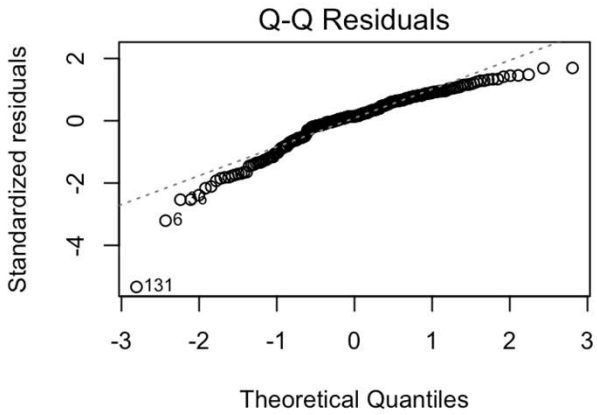
	Estimate	Std. error	*t*-statistics	*p*-value	95% CI
Intercept	2.938	0.3119	9.422	<0.0001	
TV	0.045	0.0014	32.809	0.00115	0.043,0.049
Radio	0.188	0.0086	21.893	<0.0001	0.172,0.206
Newspaper	-0.001	0.0058	-0.177	0.86	−0.013,0.011

The Marketing Plan

❻ *Is the relationship linear?*

If the relationships are linear, then the residual plots should display no pattern. In the case of the Advertising data, we observe a non-linear effect.





Moving beyond linearity...

In the next module, we will see that we can improve least squares using ridge regression, the lasso, principal component regression and other techniques. In that setting, the improvements is obtained by reducing the complexity of the linear model. But we will be still using the linear model that can be improved only so far.

Often much simpler way is to relax the assumption of the linearity whilst still attempting to retain as much interpretability.

Extensions of the Linear Model:

1. Polynomial regression.
2. Step functions and the basis function.
3. Regression splines.
4. Smoothing splines.
5. Local regression.